



FLORIDA STATE UNIVERSITY
DEPARTMENT OF SCIENTIFIC COMPUTING

WRITTEN PRELIMINARY EXAMINATION
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Question - 3

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1 Ordinary Differential Equation[Dr. Quaife]

The given system of initial value problem:

$$\frac{dL}{dt} = f_1(L, X, Y) = -4\alpha X - 4800L, \quad (1)$$

$$\frac{dX}{dt} = f_2(L, X, Y) = -5L(Y - 0.01) - 330X, \quad (2)$$

$$\frac{dY}{dt} = f_3(L, X, Y) = 5LX - 330(Y - 0.1) \quad (3)$$

1.1 Part (a)

Code for testing the solver with $\alpha = 0.1$

```

1 # Define the system of differential equations
2 def system(t, variables, alpha):
3     L, X, Y = variables
4     dL_dt = -4 * alpha * X - 4800 * L
5     dX_dt = -5 * L * (Y - 0.01) - 330 * X
6     dY_dt = 5 * L * X - 330 * (Y - 0.1)
7     return [dL_dt, dX_dt, dY_dt]
8
9 # initial conditions
10 initial_conditions = [0.01, 0.01, 0.1]
11
12 # alpha
13 alpha = 0.1
14
15 # time for which to solve the system
16 t_span = [0, 1]
17
18 # Solve the system of IVPs using solve_ivp
19 solution = solve_ivp(system, t_span, initial_conditions, args=(alpha,),
20                       method='RK45')
21
22 # Printing the solution
23 print("Time Points (t):\n", solution.t)
24 print("\nVariables (L, X, Y):")
25 for i, variable in enumerate(['\nL: ', '\nX: ', '\nY: ']):
26     print(f"{variable} {solution.y[i]}")

```

RESULTS:

Time Points (t):

[0. 0.000196 0.000392 ... 0.998839 0.999602 1.]

Dependent Variables (L, X, Y):

L: [0.01 0.003902 0.001523 ... 0.000001 0.000002 0.]

X: [0.01 0.009373 0.008785 ... 0. 0. 0.]

Y: [0.1 0.1 0.1 ... 0.1 0.1 0.1]

1.2 Part (b)

The IVP has three fixed points. If a solution reaches one of these fixed points, it stays there indefinitely.

First Fixed Point:

$$(L_1, X_1, Y_1) = (0, 0, 0.1)$$

Second and Third Fixed Points:

$$L_2 = 66\sqrt{\frac{\alpha}{880000} - 1}, \quad X_2 = -\frac{1200}{\alpha}L_2, \quad Y_2 = 0.1 - \frac{200}{11\alpha}L_2^2$$

$$L_3 = -66\sqrt{\frac{\alpha}{880000} - 1}, \quad X_3 = \frac{1200}{\alpha}L_3, \quad Y_3 = 0.1 - \frac{200}{11\alpha}L_3^2$$

For $\alpha = 1 \times 10^6$

Fixed points for alpha = 1000000.0:

Fixed Point 1: (0, 0, 0.1)

Fixed Point 2: (24.372115, -0.029247, 0.089200)

Fixed Point 3: (-24.372115, 0.029247, 0.089200)

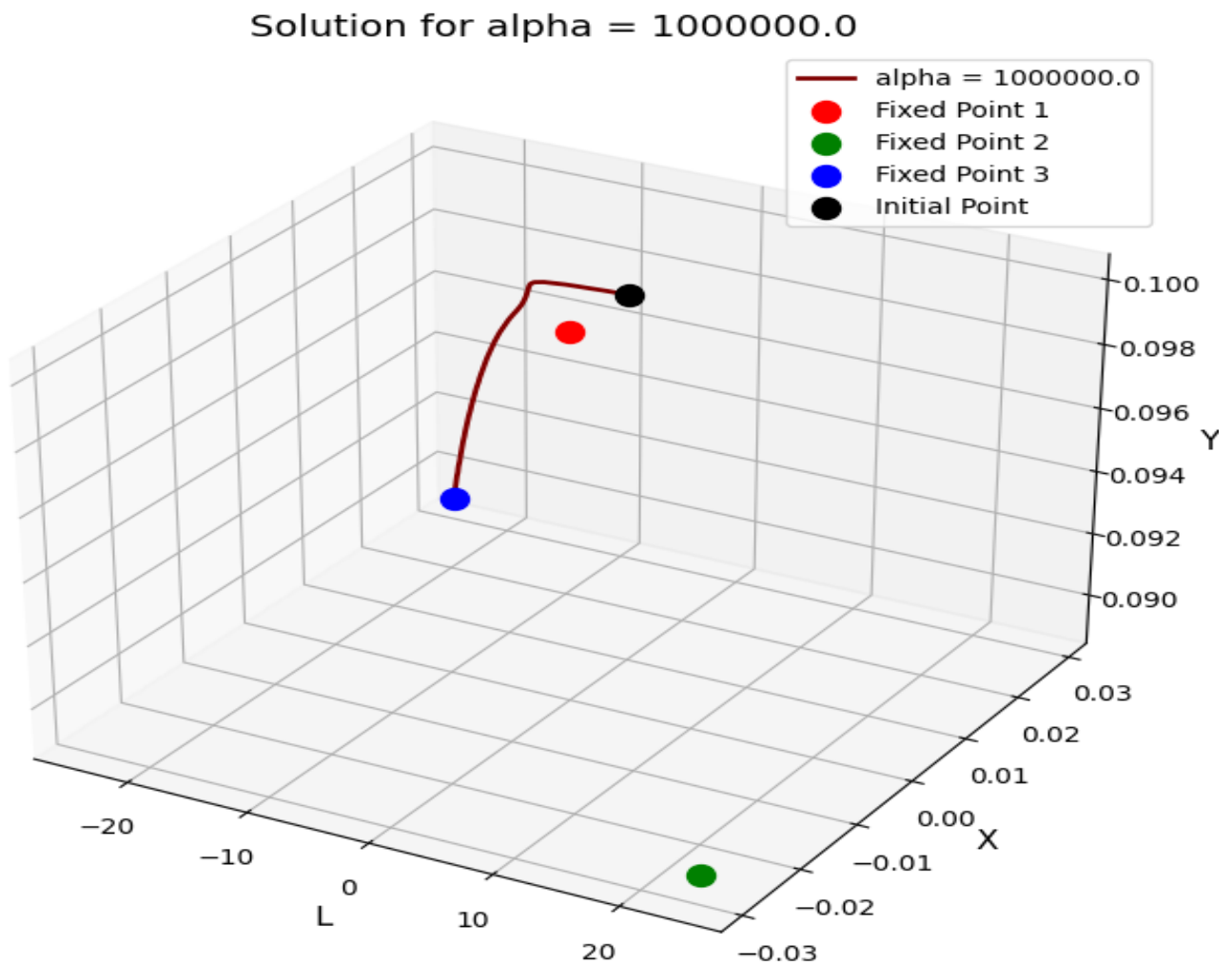


Figure 1: Solution for $\alpha = 1 \times 10^6$.

Behavior: For $\alpha = 1 \times 10^6$, the system quickly stabilizes and stays at fixed point (L_3, X_3, Y_3) . Afterwards, it remains there for the entire timespan.

For $\alpha = 3 \times 10^6$

Fixed points for alpha = 3000000.0:

Fixed Point 1: (0, 0, 0.1)

Fixed Point 2: (102.440226, -0.040976, 0.036400)

Fixed Point 3: (-102.440226, 0.040976, 0.036400)

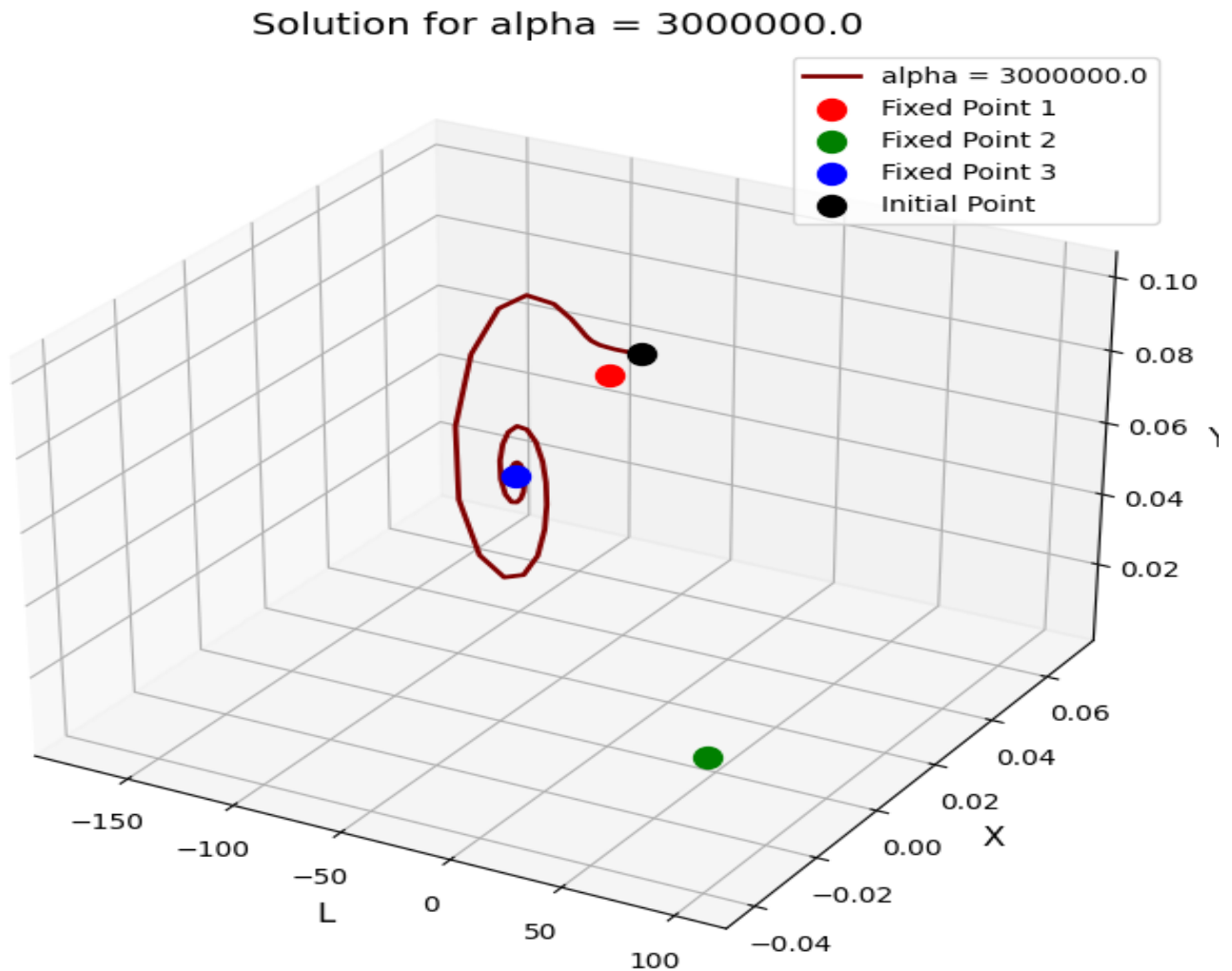


Figure 2: Solution for $\alpha = 3 \times 10^6$.

Behavior: For $\alpha = 3 \times 10^6$, the system takes some time to stabilize and then comes at fixed point (L_3, X_3, Y_3) . Afterwards, it remains there for the entire time span. The fixed points 2 and 3, in this case, are far away from the origin. Because of this the solution takes some time, oscillate and then reaches the fixed point 3.

For $\alpha = 2 \times 10^7$

Fixed points for alpha = 20000000.0:

Fixed Point 1: (0, 0, 0.1)

Fixed Point 2: (307.642650, -0.018459, 0.013960)

Fixed Point 3: (-307.642650, 0.018459, 0.013960)

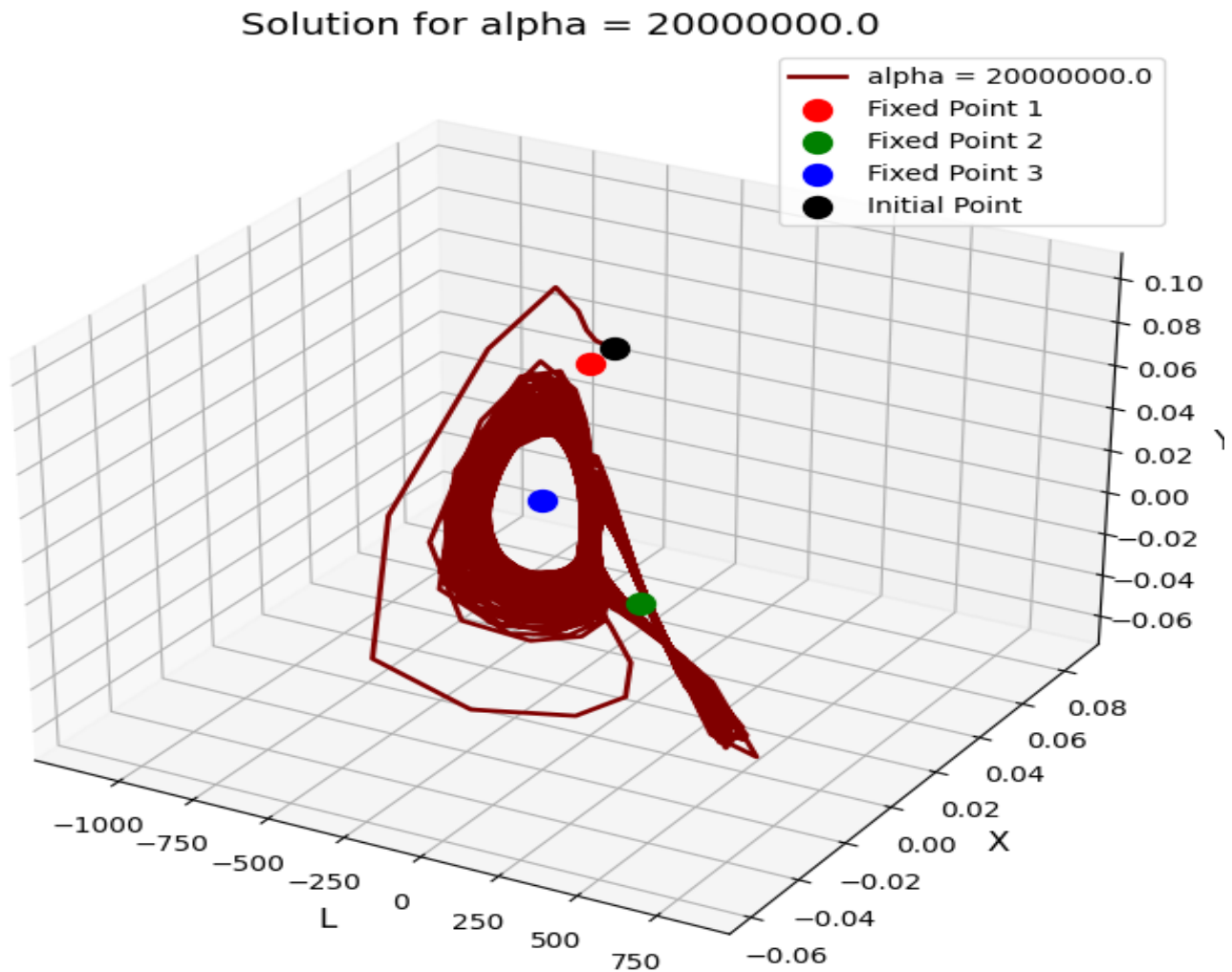


Figure 3: Solution for $\alpha = 2 \times 10^7$.

Behavior: For $\alpha = 2 \times 10^7$, the fixed points (L_2, X_2, Y_2) and (L_3, X_3, Y_3) are significantly further from the origin compared to the other two cases. The trajectories will exhibit even more oscillatory behavior with larger amplitudes. The system takes longer to stabilize and oscillates more significantly before settling near the fixed point 2 (It appears to at least! I am not sure if it is staying there because the perspective is not clear in plot).

Summary:

- As α increases, the fixed points (L_2, X_2, Y_2) and (L_3, X_3, Y_3) move further from the origin. The system exhibits more significant oscillatory behavior and takes longer to stabilize.
- For lower values of α , the system stabilizes more quickly and has lesser oscillations. For higher values, the system has larger oscillations and requires more time to approach the fixed points.

- Since the initial conditions are close to the first fixed point, the trajectories initially tend towards this point before oscillating around and eventually settles near the second or third fixed points depending on the value of α .

1.3 Part (c)

The Jacobian matrix J is given by:

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial L} & \frac{\partial f_1}{\partial X} & \frac{\partial f_1}{\partial Y} \\ \frac{\partial f_2}{\partial L} & \frac{\partial f_2}{\partial X} & \frac{\partial f_2}{\partial Y} \\ \frac{\partial f_3}{\partial L} & \frac{\partial f_3}{\partial X} & \frac{\partial f_3}{\partial Y} \end{bmatrix}$$

Using expressions for f_1 , f_2 and f_3 from eq-1, 2 and 3 respectively, we can calculate the partial derivatives as follow:

$$\frac{\partial f_1}{\partial L} = -4800, \quad \frac{\partial f_1}{\partial X} = -4\alpha, \quad \frac{\partial f_1}{\partial Y} = 0$$

$$\frac{\partial f_2}{\partial L} = -5(Y - 0.01), \quad \frac{\partial f_2}{\partial X} = -330, \quad \frac{\partial f_2}{\partial Y} = -5L$$

$$\frac{\partial f_3}{\partial L} = 5X, \quad \frac{\partial f_3}{\partial X} = 5L, \quad \frac{\partial f_3}{\partial Y} = -330$$

Hence, we have:

$$J = \begin{bmatrix} -4800 & -4\alpha & 0 \\ -5(Y - 0.01) & -330 & -5L \\ 5X & 5L & -330 \end{bmatrix}$$

1.4 Part (d)

Following is the code to define Jacobian of matrix and then compute its eigenvalues for different values of α .

```

1 # Defining the Jacobian matrix
2 def jacobian(L, X, Y, alpha):
3     J = np.array([
4         [-4800, -4*alpha, 0],
5         [-5*(Y - 0.01), -330, -5*L],
6         [5*X, 5*L, -330]
7     ])
8     return J
9
10 # Fixed points
11 fixed_points = [(L1, X1, Y1), (L2, X2, Y2), (L3, X3, Y3)]
12
13 # Values of alpha
14 alpha_values = [1e6, 3e6, 2e7]
15
16 # Loop over alpha values
17 for alpha in alpha_values:
18     print("-"*60)
19     print(f"Eigenvalues for alpha = {alpha}:")

```

```

20     print("-"*60)
21
22     for i, point in enumerate(fixed_points):
23         L, X, Y = point
24
25         # Computing the Jacobian matrix
26         J = jacobian(L, X, Y, alpha)
27
28         # Computing eigenvalues
29         eigenvalues = np.linalg.eigvals(J)
30         print(f"Fixed point {i+1}: {eigenvalues} ")
31
32     print("-"*60)
33     print("\n")

```

Eigenvalues for alpha = 1000000.0:

Fixed point 1: [-5171.76 41.76 -330.]

Fixed point 2: [-4840.73 -309.63 + 1571.69j -309.63 - 1571.69j]

Fixed point 3: [-4840.73 -309.63 + 1571.69j -309.63 - 1571.69j]

Eigenvalues for alpha = 3000000.0:

Fixed point 1: [-5789.16 659.16 -330.]

Fixed point 2: [-4919.26 -270.36 + 1634.70j -270.36 -1634.70j]

Fixed point 3: [-4919.26 -270.36 + 1634.70j -270.36 -1634.70j]

Eigenvalue for alpha = 20000000.0:

Fixed point 1: [-8967.75 3837.75 -330.]

Fixed point 2: [-5476.22 8.11 + 2036.61j 8.11 - 2036.61j]

Fixed point 3: [-5476.22 8.11 + 2036.61j 8.11 - 2036.61j]

1.5 Part (e)

For this part I have taken help from [LibreTexts](#)

Eigenvalue Type	Stability	Oscillatory Behavior	Notation
All Real and +	Unstable	None	Unstable Node
All Real and -	Stable	None	Stable Node
Mixed + & - Real	Unstable	None	Unstable Saddle Point
$+a + bi$	Unstable	Undamped	Unstable Spiral
$-a + bi$	Stable	Damped	Stable Spiral
$0 + bi$	Unstable	Undamped	Circle

Table 1: Types of Eigenvalues and Their Stability Characteristics.^[1]

For $\alpha = 1 \times 10^6$

- Fixed point 1: Eigenvalues have mixed + and - real parts, indicating instability. This is the reason that the solution never converges to this fixed point.
- Fixed points 2 and 3: They have complex eigenvalues with negative real parts, suggesting stability with damped oscillations.

For $\alpha = 3 \times 10^6$

- Fixed point 1: Eigenvalues have mixed + and - real parts, indicating instability. This is in accordance with solution obtained in part (b). The solution never converges to this point.
- Fixed points 2 and 3: Similar to $\alpha = 1 \times 10^6$, they have complex eigenvalues with negative real parts, suggesting stability with damped oscillations.

For $\alpha = 2 \times 10^7$

- Fixed point 1: Eigenvalues are again mixed and hence not stable.
- Fixed points 2 and 3: They have complex eigenvalues with small real parts, suggesting instability with comparatively lesser damped oscillations. This can be understood from Fig-2 and 3. The oscillation stays for longer in $\alpha = 2 \times 10^7$ case.

Overall, as the α value increases, the stability of the fixed points can change. The behavior of the system is also affected, with some fixed points exhibiting damped oscillations while others show undamped oscillations.

References

- [1] Allen Woolf. Using eigenvalues and eigenvectors to find stability and solve odes, 2024. Accessed on 2024-05-25.
- [2] Documentation: https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html